

Autrin Hakimi

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PROFESSIONAL PROFILE

Software engineer with production experience building automation platforms, full-stack applications, and ML-driven systems. Currently developing real-time control software, internal tooling (JavaScript/Flask), and ML inspection systems at ASM America for semiconductor manufacturing. Published researcher (ACM SIGSPATIAL 2025) experienced in distributed systems, Python, C/C++, Java, and cloud infrastructure. Strong cross-functional collaborator with end-to-end project ownership from design through production deployment.

TECHNICAL SKILLS

Languages	Python, C/C++, Java, JavaScript, TypeScript, SQL, Bash, HTML, CSS, C# (.NET)
Web & Backend	React, Python Flask, Tkinter, Node.js, REST APIs, Android, JDBC, WebSockets
Cloud & DevOps	Docker, Git, CI/CD pipelines, Linux/Unix, Microsoft Azure, Google Cloud, Snowflake
ML & Data	PyTorch, scikit-learn, Pandas, NumPy, distributed training (64-GPU), DGL, NetworkX
Databases	MySQL, SQL, NoSQL, Neo4j, Snowflake
Automation & Control	Robot workflow orchestration, vision systems, HMI/GUI, ROS, OpenCV, MoveIt
Methodologies	Agile (Scrum/Kanban), test-driven development, cross-functional collaboration

EDUCATION

B.S., Computer Science	Aug 2021 - May 2025
Iowa State University GPA: 3.70/4.0 Magna Cum Laude Dean's List	Ames, IA

PROFESSIONAL EXPERIENCE

Systems Engineer, Embedded Software	Jan 2026 - Present
ASM America	Phoenix, AZ
- Build Python-based automation platform coordinating robots, vision sensors, and semiconductor processing equipment in real-time production workflows. - Develop and deploy ML models for precision component inspection; build full-stack internal tooling with JavaScript/HTML/CSS frontend and Python Flask backend. - Implement industrial HMI/GUI systems for diagnostics, operator workflows, and serviceability across multiple equipment configurations. - Design workflow orchestration and error-handling logic for multi-step robot operations; manage project data with SQL databases. - Own end-to-end delivery from algorithm design to production deployment; collaborate cross-functionally with mechanical, electrical, and process teams.	
AI Undergraduate Researcher	Aug 2024 - Nov 2025
Iowa State University, SwAPP Lab	Remote
- Co-authored "HydroGAT" [paper, code] accepted at ACM SIGSPATIAL 2025; implemented distributed training across 64 NVIDIA A100 GPUs achieving 15x speedup. - Engineered end-to-end data pipelines (PyTorch, DGL, Pandas, NumPy) for large-scale datasets; automated workflows via Bash scripting, cutting setup time 40%.	
Software Development Intern	May 2024 - Aug 2024
BuilderTREND	Omaha, NE
- Designed and shipped a React-based job proposal template; developed full-stack features in JavaScript, TypeScript, and C# (.NET). - Migrated legacy ASP pages to React, improving scalability; collaborated in Kanban/Scrum sprints with cross-functional teams.	
Data Science Intern	May 2023 - May 2024
Alliant Energy	Remote
- Developed predictive ML models (Random Forest, SVM) in Python and SQL analyzing electricity usage for 995K customers in Snowflake. - Predicted EV ownership for customer segmentation; recognized by Director of Data Analytics for impactful business insights.	

SELECTED PROJECTS

Autonomous Robotic Chess System [code]	
Built fully autonomous system: real-time piece recognition (OpenCV), move calculation, and UR10e robot arm control (ROS, MoveIt, Python).	
ISU Pulse: Android Mobile App [code]	
Led team of 3; won Best Manager & Coder Award, ranked Top 3. Tech: Java, WebSockets, JUnit, GitLab CI/CD.	
8-Puzzle Solver: AI Search [code]	
Implemented BFS, IDS, and A* solvers; achieved 60x speedup via multiprocessing.	